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Top Skills

Embedded Software

Modeling and Simulation

Aerospace Industries

Jason Richards

PhD Researcher in Autonomous UAV Systems MEng in Aerospace Engineering Former Airbus Defence & Space GNC Intern
Luxembourg

Summary

I am a PhD Researcher at the University of Luxembourg, working at the intersection of robotics, situational awareness and control. My research focuses on Semantic Relational SLAM for autonomous UAVs, exploring how robots can build richer representations of their environments and use this understanding for navigation, planning and decision-making. My broader interests include autonomous aerial systems, semantic mapping, robotic perception and advanced autonomy for challenging environments. I graduated First-Class with an MEng in Aerospace Engineering from the University of Nottingham, specialising in Aerospace Control Systems and Avionic Systems. I developed a strong foundation in aerospace systems, software, simulation and control, with technical experience in C/C++, Python, Linux, control design, system modelling and simulation. Alongside my degree, I was a member of UoN Space Society and served as Systems Engineer for IDeT-Sat, a 3U CubeSat developed for the UKSEDS Satellite Design Competition. I contributed to software, hardware, systems integration and simulation while coordinating the technical development of the satellite. I also served as Outreach and Sponsorship Lead, securing a £10,000 Cascade Funding grant and £400 through the RS Student Grant, organising an industry tour and creating opportunities for students to gain practical space engineering experience. I also spent a year in industry as an AOCS/GNC Intern at Airbus Defence and Space, contributing to robotic development for planetary exploration rovers. This provided practical experience in spacecraft and rover dynamics, guidance, navigation and control, simulation and autonomous systems, as well as insight into complex aerospace engineering in industry. I currently serve as Competitions Team Lead at UKSEDS (UK Students for the Exploration and Development of Space), helping oversee national student space engineering competitions. I coordinate volunteers, support participants and help deliver hands-on opportunities to develop engineering and project-management skills. My career has developed around a

common theme: building increasingly capable autonomous systems for challenging environments. I am interested in bringing together software, control, perception and systems engineering to solve complex problems across robotics and space.

Experience

SnT, Interdisciplinary Centre for Security, Reliability and Trust Doctoral Researcher

November 2025 - Present (11 months)

Luxembourg

A PhD researcher focusing on making UAVs smarter and more reliable. Primarily focused for safety-critical applications where mission success is essential. My research integrates situational awareness into UAV control systems, enabling them to better understand and respond to their environments. This has included both simulation and hands-on hardware testing, with special interest in aerial manipulation: giving UAVs the ability to perceive and interact with objects using integrated sensing and control. A core aim is minimising mission failure through robust control design that stands up to real-world challenges. Further, I am working towards advancing the Technology Readiness Level of S-Graphs - developed by the ARG group at the University of Luxembourg - supporting their progression towards real-world autonomous flight operations. The overarching goal has been to develop UAV systems that are not just innovative in theory, but also reliable and applicable in practice.

UKSEDS - UK Students for the Exploration and Development of Space

2 years 2 months

Competitions Team Lead

September 2025 - Present (1 year 1 month)

As the Volunteer Team Lead for the Competitions Department at UKSEDS (UK Students for the Exploration and Development of Space), I oversee the planning and delivery of four national student competitions: In-Orbit Satellite Assembly and Manufacturing (ISAM), National Rocketry Competition (NRC), Satellite Design Competition (SDC), and Olympus Rover Trial (ORT). In this role, I manage a team of volunteers, coordinate event logistics, and support participants throughout each stage of these competitions. Our work enables students across the UK to develop essential technical and project

management skills through hands-on engineering projects. These initiatives play a key part in upskilling future generations of space engineers and providing students with practical experience to launch careers in the space sector.

Olympus Rover Trials Volunteer

August 2024 - September 2025 (1 year 2 months)

Volunteering in organising the Olympus Rover Trial (ORT) competition. A competition designed for participants to gain experience developing skills with space rover exploration.

Worked in a team to create competition documents which contain the statements of requirements and information around the competition.

Communicated with external stakeholders to discuss concepts for competitions. Collaborating to define ideas which fit within the capabilities of teams based on previous competitions.

Did onboarding with WordPress for website development to release documents to public who are interested in joining the competition.

Airbus Defence and Space

AOCSGNC

July 2023 - August 2024 (1 year 2 months)

Stevenage, England, United Kingdom

13 Month internship at Airbus Defence & Space. Investigated the characteristics of using LIDAR's for Lunar based rover exploratory missions.

- Developed a software model of a LIDAR which was integrated into an offline simulation environment.

- Designed, built, and integrated a test bench onto a rover to capture LIDAR data

- Wrote software to perform different type of LIDAR scans which would be investigated to determine best approach for navigation

- Wrote software which would take in Point Cloud and remodel to have characteristics of different LIDAR's based on vertical and horizontal FOV's and

resolution. This was to allow for investigation of how different LIDAR's may perform for rover navigation.

- Investigated the potential of using a Vector Field Histogram navigation algorithm with point clouds from LIDAR and created a development plan for how could be integrated on rovers

Nottingham Space Society

1 year 10 months

Treasurer

July 2023 - June 2024 (1 year)

Nottingham, England, United Kingdom

Treasurer for University of Nottingham's Space Society. Role included managing funds for different engineering projects, doing costing analysis and creating a breakdown of the societies funds. Further, created presentations to interact with stakeholders of the society.

Outreach and Sponsorsip Lead

September 2022 - June 2023 (10 months)

Nottingham, England, United Kingdom

Responsible for reaching out to external parties to find support for projects. Cold contacted companies from relevant sectors relevant to the projects INROS where running.

Highlights Include:

- Tour of Airbus Defence and Space
 - Team received tour of facilities
 - Given presentation on launcher interfaces, space propulsion technologies and how safety is critical in industry
 - Presented Rocket teams CDR for review
- Raised funding through RS grassroots student fund
- Organised workshop through RS grassroots
 - Oppertunity for members to develop career plans and lead techniques for self improvement

System Engineer

September 2022 - June 2023 (10 months)

Nottingham, England, United Kingdom

System Engineer for INROS satellite team. Led team in development of a 3U CubeSat concept to enter into UKSEDS Satellite Design Competition. Project Vesper was the Societies first entry into competition.

- Managed Development of OBDH software
- Managed Project Breakdown Tree and WBS
- Soldered Electronics for EPS system
- Simulated ADCS system using MATLAB Simulink
- where in top 5 teams who made it build stage and attend competition

Education

University of Nottingham

Aerospace Engineering, Aerospace, Aeronautical and Astronautical/Space Engineering · (September 2020 - August 2025)